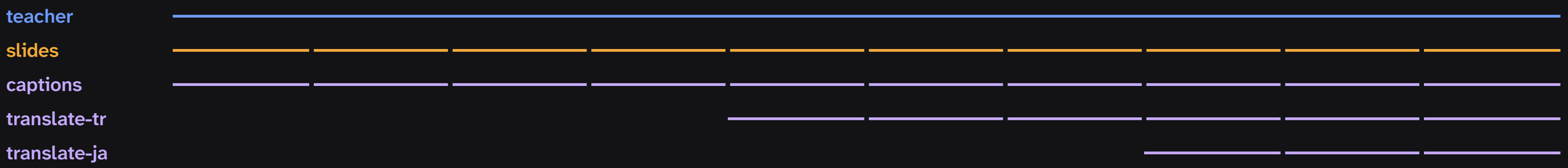
Talk Tracks

A live talk in any language, on Media over QUIC.
Why we chose MoQT, how the system runs on one small server,
and what Meta's models do in it.



A live talk reaches only part of its room.

Language

Anyone who does not share the speaker's language **gets nothing live**.

Sight

A slide nobody describes is **lost to a listener who cannot see it**.

Time

Arrive late or look away, and a live stream **has no way back**.

AI CAN CLOSE EACH GAP, BUT

captions, translation, slide descriptions and chapters, run for every talk in every language all the time, pay for output nobody reads.

SO

each AI job is a **track** in the live broadcast, and it runs **only while someone subscribes** to it.

Why MoQT and not WebRTC, HLS or WebSockets

What the talk needs	WebRTC an SFU, WHIP/WHEP	LL-HLS segments over a CDN	WebSockets a socket server	MoQT one relay, what we use
Named tracks for media and data, side by side	◐ tracks and data channels; naming is per SFU	○ one audio and one subtitle rendition at a time	◐ any bytes; the protocol is yours	● a broadcast is a set of named tracks
The publisher sees whether anyone listens	◐ depends on the SFU	○ the origin never knows who stopped	◐ you build it	● the relay subscribes upstream only on demand
Fan-out through a cache	◐ SFU cascades, no caching	● any HTTP CDN	○ every viewer is a socket on your server	● relays fan out and cache groups
Live and the past on the same track	○ real-time only	◐ a DVR window; text tied to segments	○ you build the history	● SUBSCRIBE the edge, FETCH a past group
Pushed as it is produced	● RTP and data channels	○ the player polls playlists	● pushed	● each group on its own QUIC stream
Glass-to-glass latency, p50 *	**336 ms**	**907 ms**	not measured	**570 ms**; text track **11 ms**

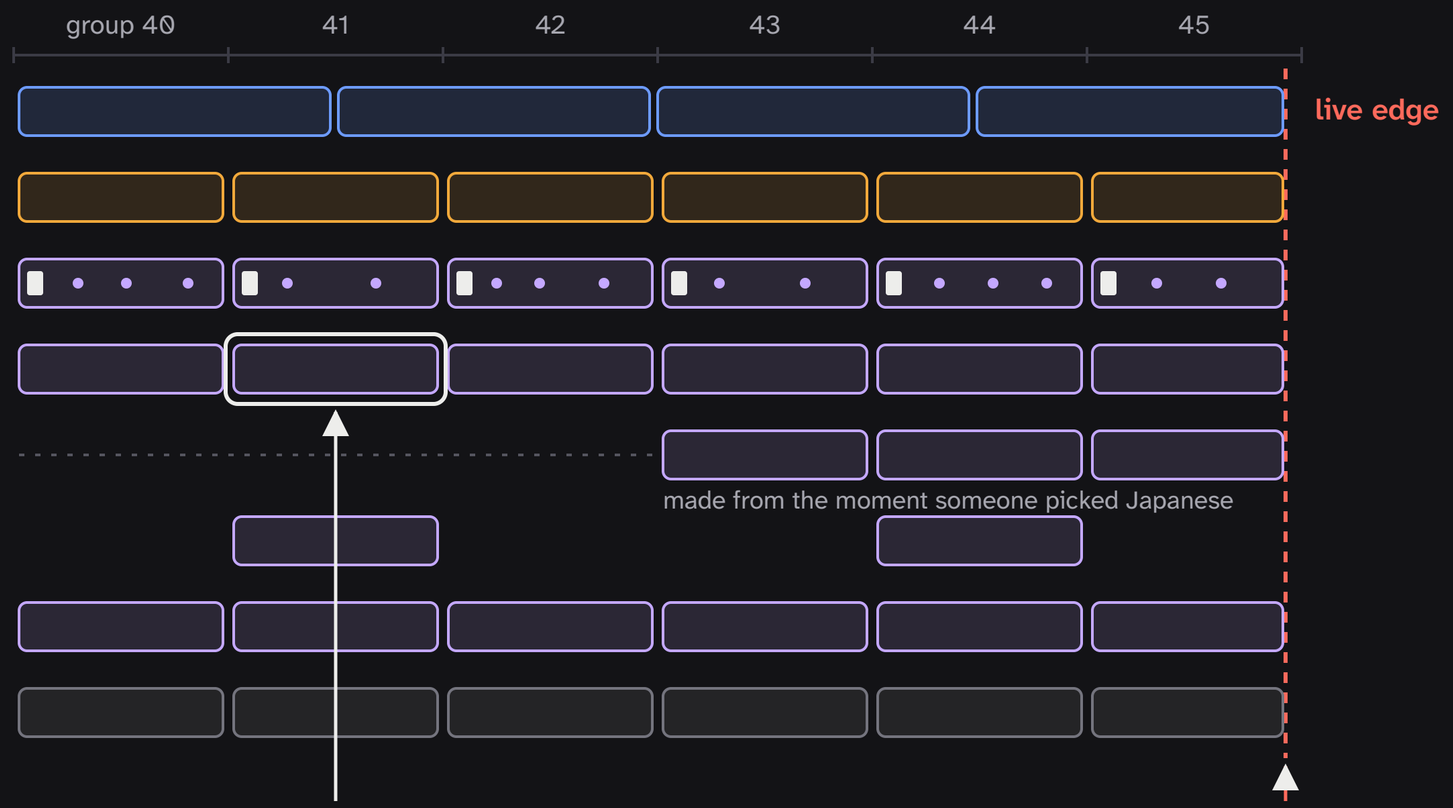
> **What we accept:** MoQT is still an IETF draft, so versions are pinned (moq-relay 0.17.8, agents on draft-18). Safari and iOS reach the relay over a WebSocket fallback. One CDN offers MoQ so far, in beta.

* One-laptop harness check over loopback, MoQ player delay 500 ms (wiki/benchmark.md). Filled = yes, half = partly, empty = no. The WebSockets column is our reading.

Media, ink and every AI job are tracks of one broadcast

talk/<room>/

- **teacher** audio, video
- **slides** deck, slide, ink, pointer
- **agents/captions**
- **agents/translate-tr**
- **agents/translate-ja**
- **agents/describe**
- **agents/chapters**
- **chat · stats** room service



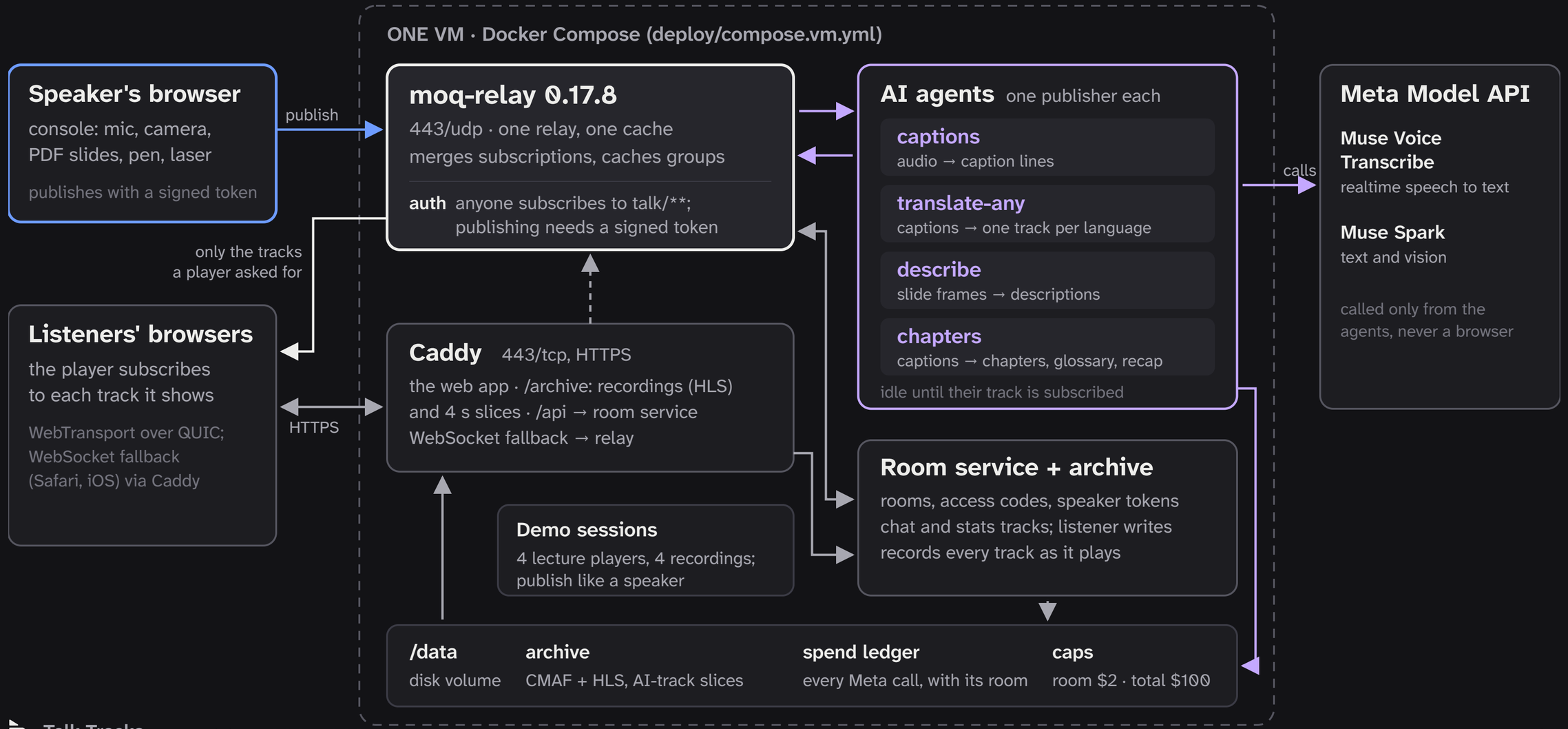
Object 0 = carry-in
the state at the slice start,
so any group is a join point.

A group = 4 s of the room clock.

Rewind: FETCH group 41
served by the relay's cache, else by the agent's archive. No new model call.

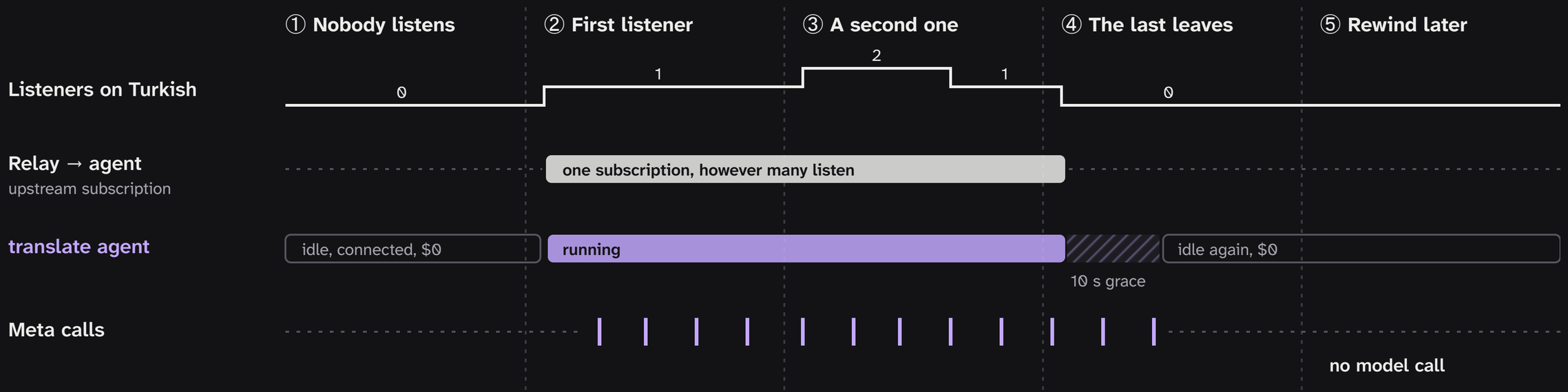
Live: SUBSCRIBE at the edge
new groups are pushed as they are made;
the relay merges every viewer into one upstream.

One relay, four agents, one small VM



Decisions 0004, 0005, 0011, 0022, 0024 · deploy/compose.vm.yml, deploy/Caddyfile

A model runs only while someone is listening



The track is announced, so a player can find it. Nothing is sent to Meta.

The relay subscribes upstream in 1–4 ms. Captions start too. First subtitle 7.2 s after the pick.

The relay merges it: nothing new upstream, no second model call per line.

Cancelled upstream in 7–12 ms (≈10 s if a tab dies). After 10 s grace the open slice closes; idle.

FETCH of a past group, from the relay's cache or the agent's archive: paid for once.

Spend ledger: each call is checked first against the room cap ($2) and the ceiling ($100). Measured: a language costs ≈ $0.025 per live minute.

Decisions 0010, 0035 · wiki/howto/lazy-agents.md (relay timings measured on moq-relay 0.17.8) · README, measured runs

Two Meta models make every AI track

Track	Model	What it does	Runs while
Captions agents/captions	**Muse Voice Transcribe** realtime, endpointing	Turns the speaker's audio into caption lines as they speak: the source text for every other AI track.	any AI track below has a listener
Translation agents/translate-<lang>	**Muse Spark** muse-spark-1.3, Standard	Translates each caption line into the language a listener picks, any of 50, with the three lines before it as context.	someone has that language on
Slide descriptions agents/describe	**Muse Spark** vision input	Describes each new slide for a listener who cannot see it; a slide shown again reuses its description.	someone has slide descriptions open
Chapters agents/chapters	**Muse Spark**	Splits the talk into chapters as it goes, explains its terms in a glossary, writes a recap for late joiners.	someone has chapters open
Recordings	**Muse Spark**	Re-punctuates a recording's caption lines without changing a word, so subtitles break on sentences.	once per recording

Every AI job is a track you choose, paid for only while someone listens.

TRY IT LIVE

talk-tracks.leventaksakal.com

THE CODE, APACHE-2.0

github.com/LeventAksakal/talk-tracks

Built from 27 Oct to 4 Nov 2026 for the Meta Global AI Developer Hackathon. Not built here, used under their licences: moq-relay and the @moq libraries, aiomoqt, Media Chrome, hls.js, PDF.js, and Meta's models.